

Frequency Domain Control Design

Ameet Deshpande

May 4, 2026

1 Time Invariant Frequency Domain Control

This article describes an approach to the control design in frequency domain. This is a promising vein of design framework, as it is independent of the state-space realization or the system order, is inherently output feedback, and most suited for systems like wind turbine, with many natural modes and disturbance which is spectral in nature.

This approach seeks to determine the optimal spectral response of the controller at all frequencies, independent from each other. It then seeks to realize such spectral response using causal transfer function or feedforward transfer functions with limited preview, using the impulse response. In case such response can not be realized exactly simultaneously for all frequencies, the best fit weighted by the disturbance spectrum is realized.

Figure 1 shows the configuration at hand. It shows C the controller and P the plant. The feedback vector is z and u and w are the control and the disturbance inputs, respectively. y_a and y_b are two vectors of objective variables, whose inner product (norm, in case two vectors are same) we wish to minimize.

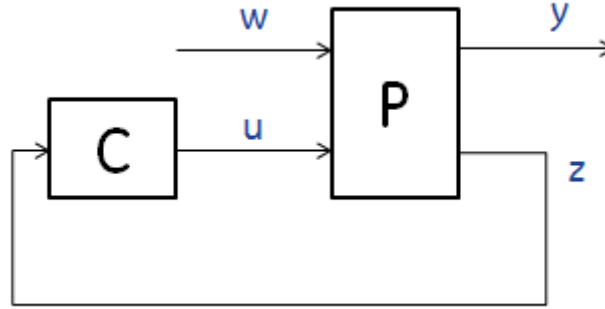


Figure 1: Schematic of the control configuration

Let us denote the corresponding s -domain transfer functions by $C(s)$, $P_{yw}(s)$, $P_{yu}(s)$, $P_{zw}(s)$ and $P_{zu}(s)$. Note that this framework allows for inclusion of the disturbance or

reference preview as well as measurement and process noise through P_{zw} . It allows the inclusion of control cost in the objective function through P_{yu} .

All the variables are corresponding complex spectra at a specific frequency, $s = i\omega$. For brevity, this dependence is omitted from the derivation below. Also note that the notation A^* denotes the complex conjugate transpose of the matrix or vector A .

Since we have $u = C(P_{zw}w + P_{zu}u)$, hence

$$u = (I - CP_{zu})^{-1}CP_{zw}w$$

Using algebra, it is easy to see the following.

$$\begin{aligned} y_a &= P_{y_a u}u + P_{y_a w}w = (P_{y_a u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_a w})w \\ y_b &= P_{y_b u}u + P_{y_b w}w = (P_{y_b u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_b w})w \end{aligned}$$

The mean 2-norm of y at a specific frequency can be expressed as

$$\begin{aligned} y_a^* y_b &= \text{Tr}(y_a^* y_b) = \text{Tr}(y_a y_b^*) \\ &= \text{Tr}((P_{y_a u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_a w})w w^* (P_{y_b u}(I - CP_{zu})^{-1}CP_{zw} + P_{y_b w})^*) \end{aligned}$$

Taking derivative with respect to the controller C , and equating it to zero, we can solve for the optimal C .

$$\frac{\partial y_a^* y_b}{\partial C} = (I_{nz} - \eta(C)) \mathcal{P}(C) \epsilon(C)$$

where

$$\begin{aligned} \eta(C) &\doteq -P_{zu}(I_{nu} - CP_{zu})^{-1}C \\ \mathcal{P}(C) &\doteq P_{zw}w w^* (P_{zw}^* C^* (I_{nu} - P_{zu}^* C^*)^{-1} (P_{y_a u}^* P_{y_b u} + P_{y_b u}^* P_{y_a u}) + (P_{y_a w}^* P_{y_b u} + P_{y_b w}^* P_{y_a u})) \\ \epsilon(C) &= (I_{nu} - CP_{zu})^{-1} \end{aligned}$$

Here $\eta(C)$ is the complementary sensitivity function and $\epsilon(C)$ is the sensitivity function. Also note that $\partial y_a^* y_b / \partial C$ is a complex gradient and actual change in $y^* y$ for a given change in δC is given by the dot product.

$$\begin{aligned} \delta(y_a^* y_b) &= \frac{\partial y_a^* y_b}{\partial C} \cdot \delta C \\ &= 0.5 \left(\frac{\partial y_a^* y_b}{\partial C} \delta C^* + \left(\frac{\partial y_a^* y_b}{\partial C} \right)^* \delta C \right) \end{aligned}$$

The optimal C_{opt} for which $\mathcal{P}(C) = 0$ and thereby $\partial y_a^* y_b / \partial C = 0$, is given by

$$C_{\text{opt}} = -X_{y_a, y_b} w w^* P_{zw}^* ((P_{zw} - P_{zu} X_{y_a, y_b}) w w^* P_{zw}^*)^{-1}$$

where $X_{y_a, y_b} \doteq (P_{y_a u}^* P_{y_b u} + P_{y_b u}^* P_{y_a u})^\dagger (P_{y_a w}^* P_{y_b u} + P_{y_b w}^* P_{y_a u})$ and $A^\dagger \doteq (A^* A)^{-1} A^*$, which is the Moore-Penrose pseudo-inverse of A .

The optimal cost using this controller is

$$\text{Tr}(y^*y)_{opt} = 4 \cdot \text{Tr}(MM^*)$$

where $M = P_{yw}P_{zw}^\dagger P_{zu}P_{yu}^\dagger P_{yw}(I - P_{zw}^\dagger P_{zu}P_{yu}^\dagger P_{yw})^{-1}w$

Future directions

- See if measurements of all states z leads to constant LQR gain controller.
- See if this can have any use in optimal sensing selection. (with coupled controller).
- See if this has any relevance/generalization in controller structure selection and structure constrained controller optimization.